

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 2.0

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Don't worry about validator error reports

- Since v2.0, this template targets the [modern PDF/UA-2 + PDF/A-4](#) accessibility standards.
- Both example PDFs pass [veraPDF](#) (strict UA-2 / PDF/A-4 conformance) cleanly.
- Several checkers (notably [Ally](#) and [PAC](#)) haven't caught up with PDF/UA-2 yet. They will report "errors" that are [false positives](#), not bugs in the templates.
- The next slide explains why the upgrade is worth doing. Per-tool symptoms and details are near the end of this deck and in `VALIDATION.md`.

Do NOT accept Ally's offered "fixes" for these findings.
They may downgrade the file to PDF 1.7, destroying PDF/UA-2 and PDF/A-4 conformance. (Verified for the title fix; other fixes are untested.)

Why v2.0 upgraded to PDF/UA-2 + PDF/A-4

What changed. v2.0 updates the `\DocumentMetadata{...}` target from the old PDF/UA-1 + PDF/A-2 (PDF 1.7) to PDF/UA-2 + PDF/A-4 (PDF 2.0).

Why this is worth doing:

- PDF/UA-2 (ISO 14289-2, 2024) is the current accessibility standard. It improves tagging for math (MathML), tables, and lists over UA-1, and is the long-term direction for accessible PDF.
- PDF/A-4 (ISO 19005-4) is the current archival standard, layered on PDF 2.0 instead of PDF 1.7. It's cleaner about deprecating legacy metadata and aligns with UA-2.
- Net effect: better screen-reader support, future-proof, and matches what modern LaTeX tagging actually produces.

The tradeoff: most accessibility checkers were written for UA-1 / PDF 1.7.

- Haven't fully caught up with new standard.
- The fix is on the checker side, not the document side.

L^AT_EX and accessibility: new requirements

- From **May 11, 2026**, course materials must meet WCAG 2.1 Level AA accessibility standards
 - Applies even to password-protected course sites (e.g., **bCourses**).
- Many of us use **L^AT_EX** to create teaching materials, both slides and documents.
- **But standard L^AT_EX (especially Beamer) does not automatically generate accessible PDFs.**
- **Good news:** The L^AT_EX Project team has developed solutions.

What is this project?

- A practical template set for accessible $\text{\LaTeX}$ using the [\$\text{\LaTeX}\$ Tagging Project](#)
 - Two document classes: articles (`article`) and slides (`ltx-talk`)
 - Two flavors of each: a `minimal` template (`minimal_slides.tex`, `minimal_article.tex`, ~40 lines, runnable in a minute) plus a `fully annotated` template (`accessible_slides.tex`, `accessible_article.tex`) that doubles as documentation
 - `Contains`: math, text, figures, and tables
- Also includes some (optional) additional slide features
 - Readable `outline slide`
 - Automatic `section separator slides`
 - Multi-page `bibliography support`
- Repo: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible $\text{\LaTeX}$: the big picture

- **Your existing $\text{\LaTeX}$ skills transfer:** mainly adding a few consistent accessibility features.
- **Most changes are local:** metadata once; then per-figure and per-table tweaks.
- **Applies to both slides & articles:**
 - Document metadata + PDF tagging
 - Alt text on images
 - Table header row tagging
 - Sufficient color contrast
 - Compile with Lua $\text{\LaTeX}$
- **Key difference:**
 - Articles: **keep your existing class** and add tagging features
 - Slides: switch to `ltx-talk` (frame syntax similar to Beamer), then recreate styling

Outline

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1. Setup

Version requirements: what actually matters

- **Engine:** must compile with [Lua \$\text{\LaTeX}\$](#) (not pdf $\text{\LaTeX}$ or Xe $\text{\LaTeX}$).
- **LaTeX kernel/format date:** must be [2025-11-01 or later](#).
- **Important:** a “TeX Live year” alone is not a reliable guarantee.
 - You need a sufficiently new $\text{\LaTeX}$ kernel *on your install*, or use Overleaf Labs.
- **Quick check (in any document preamble):**

```
\show\fmtversion
```

Overleaf (works when configured correctly)

- `ltx-talk` requires a very recent $\text{\LaTeX}$ kernel (format date 2025-11-01+).
- On Overleaf, this typically means using the Labs program (not standard Overleaf).
- **Overleaf setup (2 steps):**
 1. Join Overleaf Labs:
 - <https://www.overleaf.com/labs/participate>
 - Opt in and enable Rolling TeX Live releases
 2. Configure the project:
 - TeX Live version: Rolling TeXLive (labs) (bottom of list)
 - Compiler: Lua $\text{\LaTeX}$
- Overleaf guide: <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

2. Core requirements

Core requirement 1: document metadata

- Required for both slides and articles
- Every accessible document **must** begin with \DocumentMetadata
- Goes **before** \documentclass

```
\DocumentMetadata{
  % Request both PDF/A archival conformance and PDF/UA accessibility tagging
  pdfstandard=a-4,          % PDF/A-4 format (use a-4f only if embedding files)
  pdfstandard=ua-2,        % Add PDF/UA-2 conformance target
  lang=en-US,              % Language for screen readers
  tagging=on,              % Enable PDF tagging
  tagging-setup={
    math/setup=mathml-SE    % Enable semantic MathML generation
  }
}
```

- **Why PDF/UA-2 + PDF/A-4?** This aligns with current ISO standards and modern tagging output.

Core requirements 2–4: images, tables, and color contrast

- **Required for both slides and articles.**
- **Images:** every `\includegraphics` needs `alt={...}`.
- **Tables:** mark header rows using `\tagpdfsetup{table/header-rows={...}}`.
- **Colors:** WCAG 2.1 AA requires [4.5:1](#) contrast ratio for normal text.
- **Accessible color defaults:**

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}  
% Standard blue is fine
```

Core requirement 5: compile with Lua $\text{\LaTeX}$

- **Required for both slides and articles.**
- Compile with **Lua $\text{\LaTeX}$** (not pdf $\text{\LaTeX}$ or Xe $\text{\LaTeX}$).
- **Why Lua $\text{\LaTeX}$?**
 1. Automatic math accessibility support (no manual MathML files in normal workflows)
 2. Full tagging support
 3. Good Unicode/OpenType font support
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Editors: select LuaLaTeX as the compiler

3. Examples

Slide content: mostly the same as Beamer

- Slides live in `frame` environments (similar to Beamer).
- So your content often transfers with minimal changes.

- **Typical frame skeleton:**

```
\begin{frame}{Title}  
  \begin{itemize}  
    \item First point  
  \end{itemize}  
\end{frame}
```

- **Note (optional):** template fonts keep digits/percent/dollar signs consistent in math/text.
 - **Text:** \$1234567890%.
 - **Math:** \$1234567890%.

Figures: alt text

- Figures are included as usual (e.g., `\includegraphics`), but **add alt text**.

- Good (minimal):** `\includegraphics[alt={A capybara}]{capybara.jpg}`

- Better (more informative):**

```
\includegraphics[alt={A capybara's head facing left}]{capybara.jpg}
```

- Decorative images: use empty alt text: `alt={}`.



Caption vs. alt text: they describe different things

- A common mistake: putting the same words in `\caption{...}` and `alt={...}`.
- Different audiences, different jobs:
 - **Caption:** visible to *everyone*; an editorial label saying *why* this figure is here.
 - **Alt text:** read *instead of* the image by screen readers; a literal description of *what is depicted*.

- **Example:**

```
\includegraphics[  
  alt={A capybara's head facing left}  
{capybara.jpg}  
\caption{A capybara, used as a non-decorative illustration.}
```

- Decorative image $\Rightarrow$ `alt={}` so screen readers skip it.

Tables: header rows

- Tables use tabular as usual, but you must specify **header rows**.
- Use {1} for one header row, {1,2} for two header rows, etc.
- **Note:** PAC may still warn about header associations in some cases; see [Validator-specific notes](#).

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccrcccr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

Code listings: contrast and tagging

- For code, use a real verbatim environment, not `\texttt` inside boxes – screen readers then see the code as text.
- **Heads up:** as of TeX Live 2026, the popular syntax-highlighting packages `listings` and `minted` are **not yet compatible** with the LaTeX tagging engine; they emit untagged graphical blocks. Tracked at <https://github.com/latex3/tagging-project/issues/70> and <https://github.com/latex3/tagging-project/issues/1060>.
- The temporary working path is `fancyvrb`'s `\VerbatimInput` (used below). Monochrome, but tagged – which is what accessibility needs.

```
1 # Convert a price series to log returns.
2 import numpy as np
3
4
5 def log_returns(prices):
6     """Return  $\ln(P_t / P_{t-1})$  for a 1-D series."""
7     return np.diff(np.log(prices))
```

4. Slides versus articles

Articles vs. slides

- **Articles:** keep your existing class (e.g., `article`) and add the checklist items.
- **Slides:** switch to `ltx-talk`.
 - Your `\begin{frame}` content often stays the same.
 - Remove Beamer themes/templates; recreate styling with standard $\text{\LaTeX}$ packages (as in this template).
- **If you're converting Beamer slides:**

<code>\documentclass{ltx-talk}</code> (instead of beamer)

Getting started (new vs. converting): keep it simple

- **New documents:**

- **Try first:** `minimal_slides.tex` or `minimal_article.tex` (~40 lines each) – the bare-minimum runnable example.
- **Then copy:** `accessible_slides.tex` or `accessible_article.tex` for the fully featured template, replace content, keep the preamble structure.

- **Converting existing documents:**

- add metadata + Lua^AT_EX + alt text + table header rows
- for slides, switch to `ltx-talk` and redo styling once

- Repo: https://github.com/rhstanton/accessible_LaTeX

5. Validation and known issues

Common pitfalls

- **Confusing PDF tagging with PDF bookmarks**
 - tagging=on creates the structure tree for **screen readers** (required).
 - PDF bookmarks (navigation pane) are separate and optional.
- **Forgetting alt text**
 - Every `\includegraphics` needs `alt={...}` (or `alt={}` if decorative).
- **Not tagging table header rows**
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table.
- **Low contrast colors**
 - Avoid light colors (e.g., plain yellow/cyan).
 - Use darker variants for red/green (defined in this template).
- **Wrong engine or old kernel**
 - Use Lua $\text{\LaTeX}$ and ensure format date 2025-11-01 or later.

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- Practical workflow: run 2+ tools and compare findings.
- **Common tools:**
 - Ally (bCourses)
 - veraPDF (PDF/A conformance): <https://verapdf.org/>
 - PAC (PDF accessibility): <https://pac.pdf-accessibility.org/>
 - Adobe Acrobat (useful but has known quirks)

Optional cleanup for strict validators: `strip_af.py`

- The metadata settings produce a fully tagged PDF. Note: [Ally](#) reports some *false* errors under PDF/UA-2 – see “Validator-specific notes” for what they are and why to ignore them.
- Some strict tools (e.g., [veraPDF](#)) may still complain about /AF or embedded-file structures.
- To remove these, run:

```
python strip_af.py build/accessable_slides.pdf
```

- Prerequisite: install the Python package [pikepdf](#) first: `python -m pip install pikepdf`.
- Then validate again with veraPDF.

Validator-specific notes (reference)

New under PDF/UA-2 (already covered earlier):

- Ally reports every inline equation as “image without a description” — it doesn’t yet parse UA-2 MathML.
- Ally reports “The PDF does not have a title” — it reads the legacy PDF-1.x /Title key; **do not accept its offered “fix”**.
- PAC 26.1.0 has no UA-2 checker; it silently validates against PDF/UA-1. Failure counts are UA-1 vs UA-2 differences, not real bugs.

Older false positives (also present under UA-1):

- **Acrobat: “Lbl and LBody – Failed.”** Known Acrobat checker bug on tagged lists, captions, and section numbers. Reference: <https://tex.stackexchange.com/a/709655/2388>.
- **PAC: “Table header cell has no associated subcells.”** Reported even when table/header-rows is set correctly. Still present in PAC 26.1.0. Release notes: <https://pac.pdf-accessibility.org/en/resources/release-notes>.

Full status and per-tool details: `VALIDATION.md` in the repo.

6. (Optional) additional utilities

Optional presentation utilities

- This template includes **optional helpers** split across two packages
 - **Not required for accessibility**
 - Useful for common presentation tasks
 - Easy to customize or remove (each package is independent)
- **Two optional features:**
 1. Automatic section separator slides (in `slide_utils.sty`)
 2. Multi-page bibliography support (in `slide_bib.sty`)
- See those `.sty` files for implementation details and customization options.
- **Note:** There's additional code in the main preamble to get readable table of contents.

Optional feature 1: section separator slides

- **What it does:** Automatically inserts a separator slide after each `\section` command
 - Displays section number and title in large text
 - Centered on the slide for visual emphasis
 - Uses accessible colors (AccessibleRed for sections, blue for subsections)
- **Example:** After `\section{Core requirements}`, automatically creates a slide saying

3. Core requirements

- **To disable:** remove `\usepackage{slide_utils}` or comment out Section 1 of `slide_utils.sty`
- **To customize:** Edit colors, fonts, or spacing in the frame template

Optional feature 2: multi-page bibliography

- **Problem:** `\bibliography` inside frame environment puts all references on one slide.
 - For large bibliographies, this can be unreadable.
 - This was easy with Beamer, using the `allowframebreaks` option.
 - But `ltx-talk` does not have a similar option.
- **Solution:** Write code to split bibliography automatically across multiple slides
 - Works with standard BibTeX workflow
 - Uses Lua script ([rsbibsplitsplit.lua](#)) to split the `.bbl` file
 - (Lua^ATeX provides access to the [Lua](#) programming language)
- **Usage:** As usual (see end of this `.tex` file for complete example)
 - `\cite{key1,key2,...}` to specify references
 - `\bibliography{bibfile}` to specify bib file
 - (Set `\RS@bibmax` to customize max per slide; default=8).
 - **Note:** Do not put `\bibliography` command inside frame environment
- See `slide_bib.sty` for detailed workflow explanation.

7. Quick start

Quick start: accessible $\text{\LaTeX}$ on one slide

- **To make an accessible $\text{\LaTeX}$ PDF:**
 1. Add `\DocumentMetadata{...}` (before `\documentclass`)
 2. Compile with $\text{\Lua\LaTeX}$
 3. Every figure: `\includegraphics[alt={...}]{...}`
 4. Every table: `\tagpdfsetup{table/header-rows={...}}`
 5. Use high-contrast colors (4.5:1) and validate with 2+ tools
- Template repo: https://github.com/rhstanton/accessible_LaTeX
- Questions: richard.stanton@berkeley.edu

8. References

References

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